

B TRANSITION BETWEEN INDEPENDENT VARIABLES

This appendix details the derivation of s_{tr} (short for ‘s transition’), the s -coordinate where the transition between the independent variables s and t occurs. $s_{\text{tr}} = t_{\text{tr}}$ is defined as the specific value of s for which

$$\frac{\partial}{\partial s} = \frac{\partial}{\partial t}. \quad (\text{B.1})$$

This criterion is chosen for convenience because it implies that the state of the system, described by the state matrix X , does not change when we switch between s and t . Note that the criterion defines a unique value because for $s = s_{\text{tr}}$ we have

$$\frac{1}{\kappa} = \frac{\partial U^{0*}}{\partial t} = \frac{\partial U^{0*}}{\partial s} = \frac{1}{K^*} \quad (\text{B.2})$$

and the equation

$$K^* = \kappa \quad (\text{B.3})$$

has a unique solution because, regardless of stability, K^* is a monotonic function of s increasing from 0 at $s = 0$ to ∞ . For $s < s_{\text{tr}}$, numerical instability suggests you need to swap from s to t in the solution algorithm. To derive an equation for s_{tr} , we can start by using the chain rule to rewrite (B.1) as

$$\frac{\partial t}{\partial s} = 1. \quad (\text{B.4})$$

Now if we substitute the definitions of $t = U^{0*} \kappa = \frac{U^0}{u_*} \kappa$ and $s = kz$, we find that

$$\begin{aligned} \frac{\partial(U^{0*} \kappa)}{\partial(kz)} &= 1 \\ \frac{\kappa}{ku_*} \frac{\partial U^0}{\partial z} &= 1. \end{aligned} \quad (\text{B.5})$$

Multiplying this through by kz and substituting the definition of ϕ_m from (2.5), we can write that

$$\begin{aligned} \frac{\kappa z}{u_*} \frac{\partial U^0}{\partial z} &= kz \\ \phi_m &= s_{\text{tr}}. \end{aligned} \quad (\text{B.6})$$

For stable and neutral conditions ($\zeta \equiv z/L \geq 0$), there is a straightforward solution for s_{tr} ,

$$\begin{aligned} kz_c &= 1 + 5 \frac{kz_c}{kL} \\ kz_c &= \frac{1}{1 - \frac{5}{kL}}. \end{aligned} \quad (\text{B.7})$$

For unstable conditions ($\zeta < 0$), the solution is a little more involved. We start by substituting ϕ_m and rearranging to get the quintic equation

$$\begin{aligned} kz_c &= \left(1 - 19.3 \frac{kz_c}{kL}\right)^{-1/4} \\ (kz_c)^{-4} &= 1 - 19.3 \frac{kz_c}{kL} \\ -\frac{19.3}{kL} (kz_c)^5 + (kz_c)^4 - 1 &= 0 \\ -\frac{19.3}{kL} s_{\text{tr}}^5 + s_{\text{tr}}^4 - 1 &= 0. \end{aligned} \quad (\text{B.8})$$

This can be solved numerically using the Newton-Raphson method, which states that

$$s_{n+1} = s_n - \frac{f(s_n)}{f'(s_n)}, \quad (\text{B.9})$$

where s_{n+1} is a better approximation of s_{tr} than s_n . This process is repeated until the relative change in s ,

$$\left| \frac{\Delta s}{s_n} \right| = \left| \frac{s_{n+1} - s_n}{s_n} \right| = \left| -\frac{f(s_n)}{f'(s_n)} \right| = \left| \frac{f(s_n)}{f'(s_n)} \right|, \quad (\text{B.10})$$

is less than a certain threshold, chosen according to the desired accuracy of the solution. In this case,

$$\begin{aligned} f(s_n) &= -\frac{19.3}{kL} s_n^5 + s_n^4 - 1 \\ f'(s_n) &= 5 \times -\frac{19.3}{kL} s_n^4 + 4s_n^3 \\ s_{n+1} &= s_n - \frac{-\frac{19.3}{kL} s_n^5 + s_n^4 - 1}{5 \times -\frac{19.3}{kL} s_n^4 + 4s_n^3}. \end{aligned} \quad (\text{B.11})$$

We can also say something about what a suitable initial guess might be for s_{tr} depending on the value of $-\frac{19.3}{kL}$ (also known as `cdivkL` in PyFuga). If we rewrite (B.8) as

$$s_{\text{tr}}^4 \left(-\frac{19.3}{kL} s_{\text{tr}} + 1 \right) = 1 \quad (\text{B.12})$$

then for $-\frac{19.3}{kL} \ll 1$,

$$s_{\text{tr}}^4 \left(-\frac{19.3}{kL} s_{\text{tr}} + 1 \right) \approx s_{\text{tr}}^4 = 1, \quad \Rightarrow \quad s_{\text{tr, initial}} = 1. \quad (\text{B.13})$$

Conversely, for $-\frac{19.3}{kL} \gg 1$,

$$s_{\text{tr}}^4 \left(-\frac{19.3}{kL} s_{\text{tr}} + 1 \right) \approx -\frac{19.3}{kL} s_{\text{tr}}^5 = 1, \quad \Rightarrow \quad s_{\text{tr, initial}} = \left(-\frac{19.3}{kL} \right)^{-1/5}. \quad (\text{B.14})$$

In order to cover the whole domain of values for $-\frac{19.3}{kL}$, this is implemented in PyFuga as

$$s_{\text{tr, initial}} = \begin{cases} 1 & \text{for } -\frac{19.3}{kL} > 1 \\ \left(-\frac{19.3}{kL} \right)^{-1/5} & \text{for } -\frac{19.3}{kL} \leq 1. \end{cases} \quad (\text{B.15})$$